MIL-C-11080A(MU)
30 December 1968
SUPERSEDING
MIL-C-11080(CmlC)
16 April 1951

MILITARY SPECIFICATION

COATING, CORROSION-RESISTANT

1. SCOPE

1.1 This specification covers one type of an aluminum corrosion-resistant coating.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-A-250/8	-	Aluminum Alloy, 5052, Plate and Sheet.
QQ-S-781	-	Steel Strapping, Flat.
TT-E-515	-	Enamel, Alkyd, Lustreless, Quick-Drying.
TT-P-143	-	Paints, Varnish, Lacquer, and Related Materials; Packaging, Packing and Marking.
TT-W-00572	-	Wood Preservative; Water-Repellent.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Box, Fiberboard.

MILITARY

MIL-C-5541	-	Chemical Films and Chemical Film Materials, for Aluminum and Aluminum Alloys.
MIL-A-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys.

FSC 8030

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STANDARDS

FEDERAL

FED-STD-141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

D150-2-1 - Marking Diagram for Shipping Containers.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The material shall consist of either a baked or air-dried single or two-coat system, applied by dipping, spraying, or both. When a thinner, specially formulated for the coating furnished, is required, it shall be supplied by the manufacturer.

3.2 Physical requirements. The coating system shall conform to the requirements specified in table I when tested as specified in table II.

3.3 Workmanship. The finished coating shall be smooth, uniform, and free from imperfections.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

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Table I. Requirements

Test	Requirement
Color and appearance	Smooth finish that is not water-white.
Drying time 1/	Set to touch in 1 hour and dry hard in 6 hours.
Storage stability	
Full can (sealed)	No change in appearance when stored 3 months.
3/4-full glass jar (sealed)	No change in appearance when stored for 30 days; except for a continuous, readily removable surface skin not more than 1/16-inch thick.
Immersion resistance to hydrochloric acid	No deterioration in 96 hours.
Hardness	Average of not less than 65 Sward units for 10 determinations.
Knife	No chipping.
Nonvolatile content	33 percent (min.).
Compatibility	Compatible with enamel.

1/ Only applicable to air-drying cotaings.

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Table II. Methods of test

Test	Method 1/	Remarks
Color and appearance		Visually examine prepared test panels (see 4.2.4.1).
Drying time	4061.1	Film thickness shall be as specified in 4.2.4.1.
Storage stability Full can and 3/4-full glass jar	3022	Storage temperature 72° ± 5°F.
Immersion resistance to hydrochloric acid	6011	The immersion fluid shall be a 10-percent hydrochloric acid solution (sp. gr. 1.05). One half of the panels tested shall be creased with a longitudinal crease applied by constant pressure over a rib of 0.030 ± 0.003-inch radius and a mating die with a 0.050 ± 0.010-inch concave radius. The edges of the test panels shall be coated with paraffin.
Hardness		Use the Sward Rocker Method (see 6.3).
Knife		Cross hatch, an area with lines scribed 1/32 inch apart and visually examine for chipping.
Nonvolatile content	4042	
Compatibility with lustreless enamel		Duplicate the above tests, excepting storage stability, on test panels that have been overcoated with lustreless enamel conforming to TT-E-515.

1/ All methods referenced in table II are contained in FED-STD-141.

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4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. Each batch shall constitute a lot (see 6.4).

4.2.2 Sampling. Sampling shall be conducted in accordance with method 1021 of FED-STD-141.

4.2.3 Inspection procedure. Each sample shall be tested in accordance with table II and inspected for compliance with table I.

4.2.4 Tests.

4.2.4.1 Preparation of test panels. Prepare panels for all tests listed in table II in accordance with method 2013.1 of FED-STD-141. The prepared panels shall measure 3 by 6 by 0.016 ± 0.004 inches, shall conform to the requirements for half-hard sheets as specified in QQ-A-250/8 (H24 or H34), and shall be anodically or chemically treated in accordance with MIL-A-8625 or MIL-C-5541 respectively. The panels shall then be coated, as recommended by the manufacturer, with the test coat either by dipping or spraying in accordance with method 4341.1 or 4331.1 of FED-STD-141. Determine the film thickness of the prepared panels in accordance with method 6181 of FED-STD-141 and use, for testing, only panels having a film thickness of 0.0005 to 0.0010 inch.

4.2.5 Acceptance/rejection criteria. If any sample, taken as specified in 4.2.2, fails to meet any requirement of this specification the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level A and B. The coating material shall be packaged in either 1 or 5 gallon metal containers as specified in 3.8.1 of TT-P-143.

5.1.1 Packaging, level C. The coating material shall be packaged to provide adequate protection against deterioration and damage from the supply source to the first receiving activity for immediate use.

5.2 Packing, level A. The coating material, packaged as specified in 5.1, shall be packed in a close fitting overseas type nailed wood, wirebound, or cleated plywood box, conforming to PPP-B-621, PPP-B-585, or PPP-B-601, respectively. The boxes shall be closed and strapped in accordance with the appendix thereto. Strapping confirming to type I, class B of QQ-S-781 shall be used. The box and the finished wood parts thereof, shall be completely immersed for a minimum of three minutes in a water-repellent wood preservative conforming to composition A of TT-W-00572.

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5.2.1 Packing, level B. The coating material packaged as specified in 5.1, shall be packed in a style RSC, fiberboard box conforming to V3c of PPP-B-636. The box shall be closed and sealed in accordance with the appendix thereto.

5.2.2 Packing, level C. The coating material shall be packed the same as that specified in 5.2.1.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages shall be marked in accordance with MIL-STD-129. Each metal container shall be marked either by labeling, direct printing, or other suitable process with complete instructions for application. Shipping containers shall be marked in accordance with Drawing D150-2-1. All markings on the shipping container shall be clear, legible, and permanent after the application of the wood preservative as specified in 5.2.

6. NOTES

6.1 Intended use. This coating is intended for use as a corrosion-resistant coating for chemically treated, aluminum, protective mask canisters.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Level of packaging and packing.
c. Size of container desired.

6.3 Sward rocker method. A detailed description of the Sward Rocker Method of hardness testing is contained in the eleventh edition of "Physical and Chemical Examination of Paints, Varnishes, Lacquers and Colors", published by the Henry A. Gardner Laboratories.

6.4 Batch. A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

6.5 Suggested coatings. The following formula and coatings have been tested and conform to this specification:

1. Formula:

Ingredients	Parts by weight
Vinyl resin	11.75
Phenolic resin	117.00
Butanol	240.00
Toluene	14.00

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2. Coatings:

a. Jones and Dabney - Syntex Coating No. 9264.
b. Pittsburgh Plate Glass Co. - Synthetic Clear VD-4832.

The above information is supplied to potential suppliers with the distinct understanding that the Government makes no warranty that a coating prepared by the above formulation will conform to this specification and that irrespective of the formulation used by the supplier the requirements specified herein must be met prior to acceptance.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 8030-A010

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TO DETACH FORM - CUT ON THIS LINE

SPECIFICATION ANALYSIS SHEET	Form Approved Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 56 1426 REPLACES EDITION OF 1 OCT 54 WHICH MAY BE USED.

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Edgewood Arsenal
ATTN: SMUEA-QAES
Edgewood Arsenal, Maryland 21010

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